

# TypeScript

## 1.What is TypeScript?

TypeScript is a programming language that adds types to JavaScript so you can catch errors early while writing code, not after running it.

### Step 1: Install TypeScript

Run this command in Terminal:

```
npm install -g typescript
```

-g = install globally (whole system)

then some error occurs so write

```
sudo npm install -g typescript
```

### Step 2: Check TypeScript installed or not

```
tsc -v
```

If version shows → TypeScript installed successfully

### Step 3: Use TypeScript

Create a file:

```
touch one.ts
```

Compile TS to JS:

```
tsc one.ts
```

This will create:

```
one.ts → one.js
```

## 2.tsc one.ts --target es2016

tsc → TypeScript compiler

one.ts → TypeScript file you want to compile

--target es2016 →

“Convert my TypeScript code into JavaScript that follows ES2016 standard”

This command converts one.ts into one.js, using ES2016 JavaScript features.

### Why --target is needed?

Different browsers understand different JavaScript versions.

Examples:

es5 → very old browsers

es6 / es2015 → modern browsers

es2016 → modern + some new features

### 3.tsc --init

tsc --init creates a file called tsconfig.json

This file tells TypeScript how to compile your project.

### What happens when you run it?

After running:

tsc --init

### You will see:

tsconfig.json created

Why tsconfig.json is important?

It saves all settings in one place so you don't need to write long commands again.

Instead of this every time :

tsc one.ts --target es2016

You can just do this

tsc

### What is inside tsconfig.json? (simple)

```
{  
  
  "compilerOptions": {  
  
    "target": "es2016",  
  
    "module": "commonjs",
```

```
"strict": true  
  
}  
  
}
```

### Meaning:

target → which JavaScript version

module → how files are imported/exported

strict → strict error checking

How to use after --init

1.Create .ts files

2.Run:

**tsc**

All .ts files compile automatically

**4.tsc --watch** means TypeScript keeps watching your files and automatically recompiles them whenever you save a change.

then control c for exit

### 5.Why we need TypeScript?

We need TypeScript because it helps us find mistakes early and makes JavaScript code safer and easier to manage, especially in big projects.

- Catches errors while writing code (before running)
- Forces correct data types (number, string, boolean)
- Makes code more readable
- Easier to maintain large code
- Better support in editors (auto-suggest, hints)